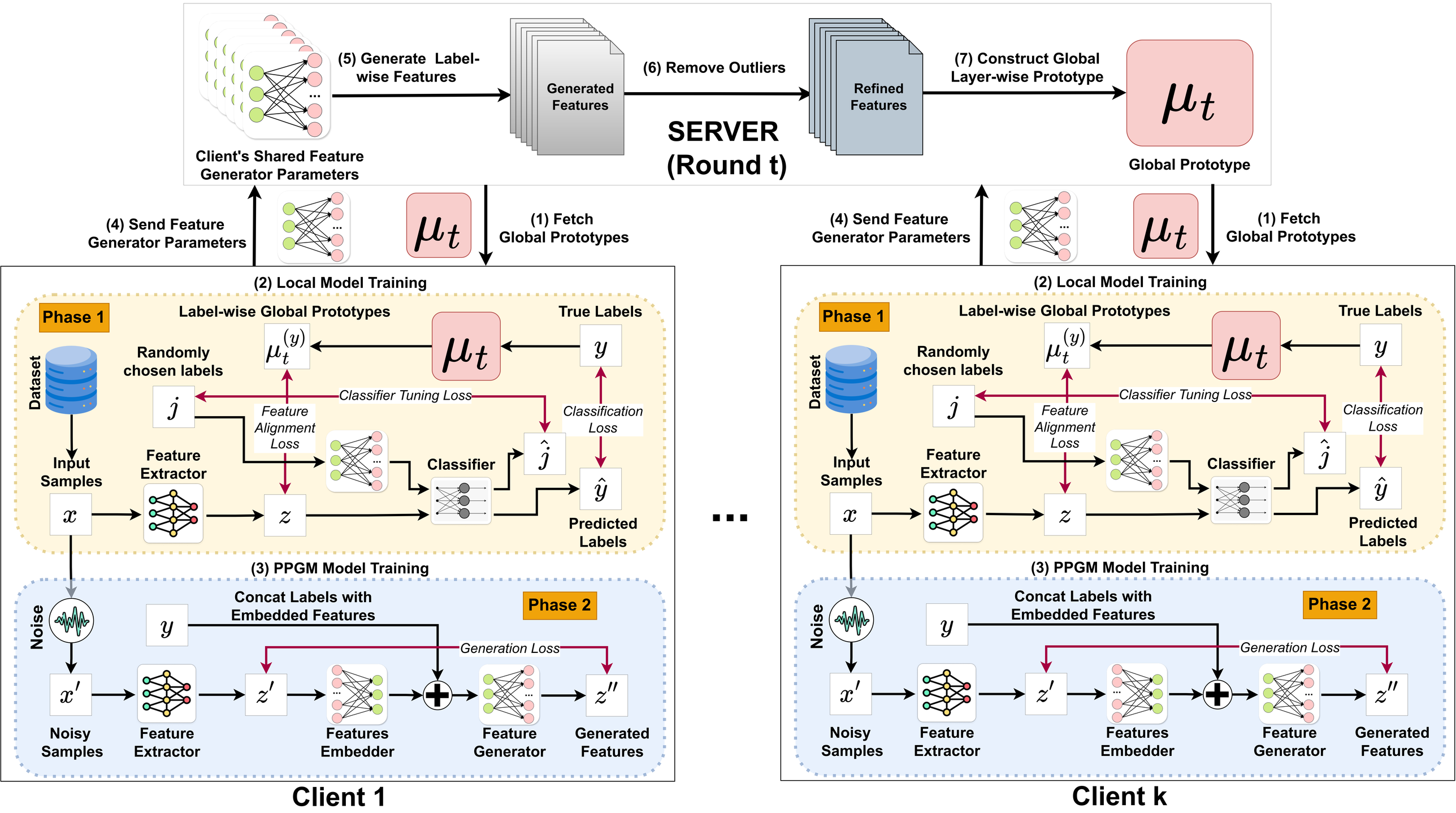

Client's Shared Feature Generator Parameters
(5) Generate Label-wise Features
Generated Features
(6) Remove Outliers
Refined Features
SERVER (Round t)
(7) Construct Global Layer-wise Prototype
Global Prototype
(4) Send Feature Generator Parameters
(1) Fetch Global Prototypes
(2) Local Model Training
Phase 1
Label-wise Global Prototypes
True Labels
Dataset
Randomly chosen labels
Classifier Tuning Loss
Feature Alignment Loss
Classification Loss
Input Samples
Feature Extractor
Classifier
Predicted Labels
(3) PPGM Model Training
Noise
Concat Labels with Embedded Features
Phase 2
Generation Loss
Noisy Samples
Feature Extractor
Features Embedder
Feature Generator
Generated Features
Client 1
Client k